

Name: _____

Period: _____

Date: _____

Topic 0.3 Guided Notes: Correlation and Experimentation

Unit 0: Science Practices | AP Psychology

Directions: Read through the topic overview page below and fill in each blank as you go. Use the tables and activities to check your understanding, then head to the quiz link at the end to test yourself.

Source page: appsychreviewhub.com/overviews/topic0.3

Word Bank

<i>Experimental Group</i>	<i>Placebo Effect</i>	<i>Correlation</i>	<i>Single-Blind Procedure</i>
<i>Correlational Study</i>	<i>Directionality Problem</i>	<i>Operational Definition</i>	<i>Variable</i>
<i>Correlation Coefficient</i>	<i>Confounding Variable</i>	<i>Regression Toward the Mean</i>	<i>Independent Variable</i>
<i>Double-Blind Procedure</i>	<i>Causation</i>	<i>Illusory Correlation</i>	<i>Placebo</i>
<i>Experiment</i>	<i>Dependent Variable</i>	<i>Control Group</i>	<i>Positive Correlation</i>
<i>Third Variable Problem</i>	<i>Random Assignment</i>	<i>Scatterplot</i>	<i>Negative Correlation</i>
<i>Random Sample</i>	<i>Falsifiability</i>		

1. The Big Picture: Tracking Variables and Finding Causes

A _____ is anything that can vary and is feasible and ethical to measure. Research methods broadly fall into two categories: Non-Experimental Methodology, where we simply observe variables to see if they are related, and experiments, where we actively manipulate variables to prove cause and effect.

2. Predicting Behavior: Correlation

A _____ is a measure of the extent to which two variables change together. Researchers plot this relationship on a _____, a graphed cluster of dots where the slope suggests the direction of the relationship.

- _____: two variables move in the same direction (as study hours increase, GPA increases).
- _____: two variables move in opposite directions (as absences increase, GPA decreases).

Researchers measure the strength of a relationship using a _____ (an r value) ranging from -1.0 to +1.0. The closer to 1.0 in either direction, the stronger the relationship.

The Two Traps of Correlation

- _____: even if A and B are correlated, we don't know if A causes B, or if B causes A.
- _____: an unmeasured, outside variable might be responsible for the observed relationship between two other variables (e.g. ice cream sales and sunburns are both driven by hot summer weather).

3. Illusions and Regressions

An _____ occurs when we believe a relationship exists between two things, but in reality, there is none — like believing people act strangely during a full moon. _____ is the tendency for extreme or unusual scores to naturally fall back toward their average over time, which can easily be mistaken for a real relationship if we're not careful.

4. Isolating Cause and Effect: The Experiment

An _____ is a research method in which an investigator actively manipulates one or more factors to observe the effect on some behavior or mental process.

The Anatomy of an Experiment

- _____ (IV): the factor that the experimenter manipulates or changes — the presumed "cause."
- _____ (DV): the outcome factor that is measured to see if it changed in response to the IV — the presumed "effect."
- _____: the group of participants exposed to the treatment.
- _____: the group that is NOT exposed to the treatment, serving as a baseline for comparison.

Apply It: The Energy Drink Study

A psychologist tests if an energy drink improves exam scores. 50 students get a 16 oz energy drink before the test; 50 students get 16 oz of plain water before the test. Both groups' exam scores (0-100%) are recorded. Fill in each part of the design.

Component	What it is in this study
Independent Variable	
Dependent Variable	
Experimental Group	
Control Group	

Controlling the Environment

Researchers must eliminate _____s — outside factors that might accidentally produce an effect. The strongest tool against this is _____ — assigning participants to groups by chance so pre-existing traits are balanced out.

Experiments must also account for the _____ — real changes a participant experiences simply because they expect the treatment to work. To control for this, the control group is often given a _____ (a fake treatment). Researchers use a _____ (only participants are unaware of their group) or the gold-standard _____ (neither participants nor researchers know) to prevent expectations from ruining the data.

Finally, a hypothesis must possess _____ — it must be structured so it's capable of being proven false by the data.

5. AP Science Practice Focus: Methods and Conclusions

Practice 2.A asks you to identify whether a study is correlational or experimental. Practice 2.B asks you to identify the Independent Variable, Dependent Variable, and any Confounding Variables, and to evaluate whether the hypothesis is falsifiable. Practice 2.C asks you to evaluate conclusions — if the study is correlational, you must point out that the researcher cannot claim causation due to third variables.

The AAQ Exam Connection

Part A asks you to state whether the study is a _____ or an experiment. Part B asks you to identify the Independent or Dependent Variable and its _____. Part F asks you to look at a scatterplot or correlation coefficient and explain how it supports or refutes the researcher's claim.



Video Notes: Psychological Research (Crash Course Psychology #2)

This video covers several research-method topics from earlier in the unit as well as this one — only answer the questions below, which focus on the parts that match this topic (correlation and experimental design).

- According to the video, why can't a correlational study prove that one thing causes another?
- In the video's maze/caffeine experiment example, what was the independent variable, and what was the dependent variable?
- Why does an experiment need a control group instead of just testing one group and seeing what happens?

6. Don't Trip Up! (Common Misconceptions)

Fill in the blank to complete each warning.

Correlation does NOT equal _____ — the only way to prove causation is through a controlled experiment.

_____ happens first — it's how you gather people from the population to join your study.

_____ happens second — it's how you divide that sample into the experimental and control groups.

7. Quick Check: Review Questions

Circle the best answer for each question below.

1. A researcher finds that as the number of hours teenagers spend on social media increases, their reported hours of sleep decreases. This relationship is best described as a:
 - A. Confounding Variable
 - B. Negative Correlation
 - C. Positive Correlation
 - D. Illusory Correlation
2. Data show that towns with more churches also tend to have more bars per capita. A researcher wrongly concludes that religion causes people to drink more. What is the researcher most likely overlooking?
 - A. The Placebo Effect
 - B. Random Assignment
 - C. The Third Variable Problem (population size affects both)
 - D. Regression Toward the Mean
3. A gambler notices that he tends to win more when he wears his "lucky hat" and becomes convinced the hat is bringing him good luck, even though he ignores the many times he wore the hat and lost. This is an example of:
 - A. Illusory Correlation
 - B. The Directionality Problem
 - C. Regression Toward the Mean
 - D. Falsifiability
4. A psychologist wants to know if a new mindfulness app reduces anxiety. She randomly assigns 60 participants to either use the app daily for a month or do nothing different, then measures each participant's anxiety score at the end of the month. In this experiment, the dependent variable is:
 - A. The participants' anxiety scores at the end of the month
 - B. Whether participants used the mindfulness app
 - C. The number of participants in each group
 - D. The length of the study (one month)
5. In a study testing a new pain reliever, neither the participants nor the researchers administering the pills know who is receiving the real medication versus a fake, inactive pill until after the results are recorded. This procedure is called a:
 - A. Double-Blind Procedure
 - B. Naturalistic Observation
 - C. Meta-Analysis
 - D. Single-Blind Procedure



Check Your Progress

Done with your notes? Head over to the [Unit 0 Quiz](https://appspsychreviewhub.com/quiz) to test what you've learned:

<https://appspsychreviewhub.com/quiz>